COMP3055
Machine Learning

Topic 7 – Data Clustering

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Supervised VS Unsupervised Learning

有标签数据，学习输入-->输出的映射函数。目标：根据样例预测新输入的正确输出。例：房价预测、图像分类。

- Supervised learning
  - Learns a function that maps an input to an output based on example input-output pairs.
  - Training data is labeled.
- Unsupervised learning

无标签数据，从数据本身发现结构或关系。目标：自动分组或降维，不依赖人工标签。例：数据聚类（如 K-Means）、PCA。

  - Learns from test data that has not been labeled.
  - Learns relationships between elements in a data set and classify the raw data without "help.”
  - Typical application includes data clustering. 数据聚类

Motivating Problems

- A true colour image – 24bits/pixel, R – 8 bits, G – 8 bits, B – 8 bits

16777216 possible colours

- A gif image - 8bits/pixel

256 possible colours

颜色表（Palette）就是一种**聚类结果**，每个代表色是一个簇的中心。
压缩思路：
原始像素 --> 找最近的代表色 --> 存索引解码--> 用索引查颜色表--> 重建图像
这种技术叫 **Indexed Colour Image**（索引色图像），常用于 GIF、PNG 等格式。

Motivating Problems

(r, g, b)

For each pixel in the original image

Find the closest colour in the Colour Table

r_0	g_0	b_0
r_1	g_1	b_1
⋮		
r_{255}	g_{255}	b_{255}

Colour Table

Record the index of that colour (for storage or transmission)

To reconstruct the image, place the indexed colour from the Colour Table at the corresponding spatial location

🧩 **左边：原始图像中的像素**

- 每个像素有一个颜色，用三元组 (r,g,b) 表示。
  例如：一个红点的颜色值可能是 (120, 80, 200)。
- **绿色框：**
  “For each pixel in the original image, find the closest colour in the Colour Table.”
  意思是：
  👉 对每个像素，在颜色表（只有256个颜色）里找到**最接近的颜色。**
  (“接近”通常用欧式距离衡量，如 $\sqrt{(r-r_i)^2+(g-g_i)^2+(b-b_i)^2}$)

把所有像素颜色分成几组（聚类），每组用一个代表色；
原图只需存代表色编号，就能大幅压缩图像。

Motivating Problems

(r, g, b)

r_0	g_0	b_0
r_1	g_1	b_1
⋮		
r_{255}	g_{255}	b_2

Colour Table

For each pixel in the original image

Find the closest colour in the Colour Table

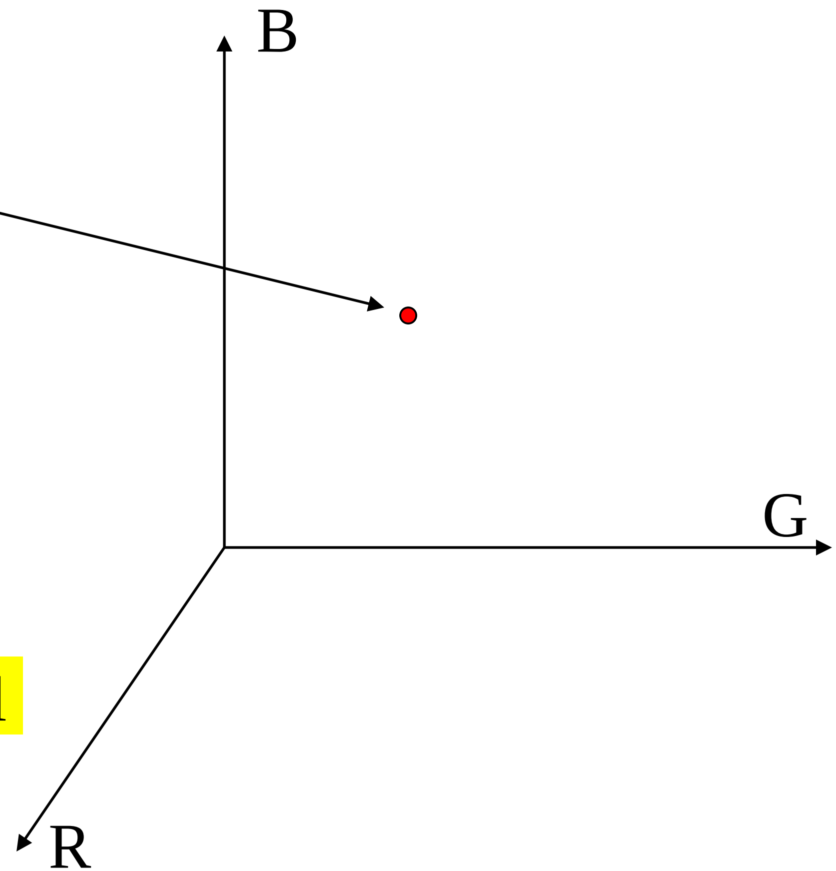
How to choose the colours in the table?

聚类算法：K-Means

the
indexed colour from the Colour Table at the corresponding spatial location

Motivating Problems

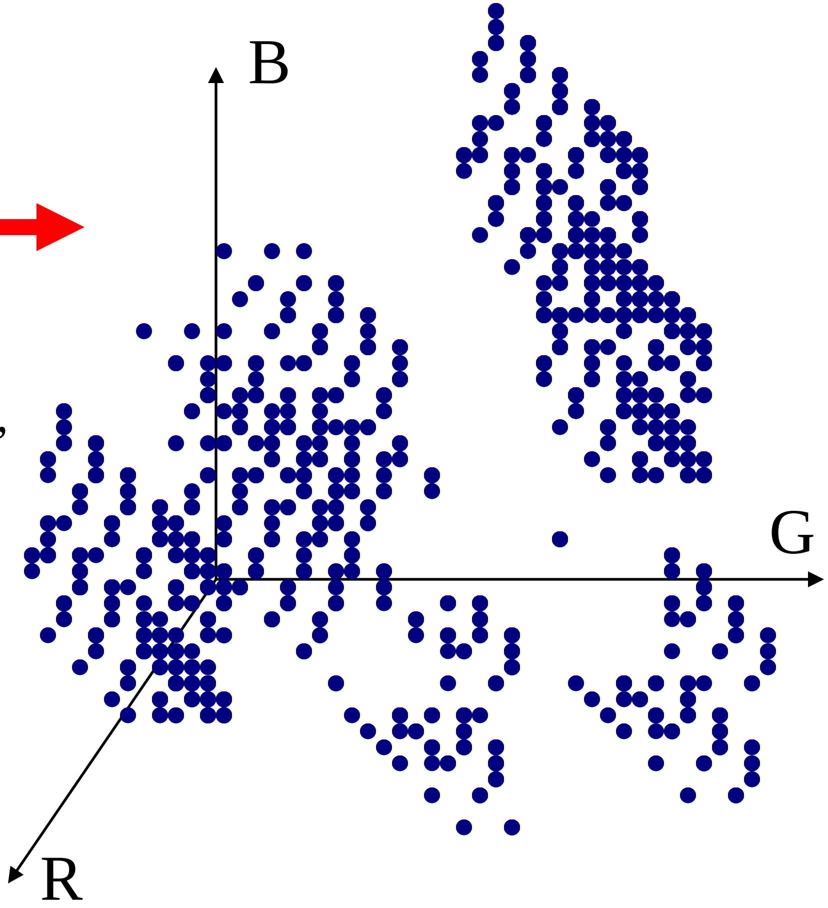
(r, g, b)



A pixel corresponding to
A point in the 3 dimensional
R, G, B space

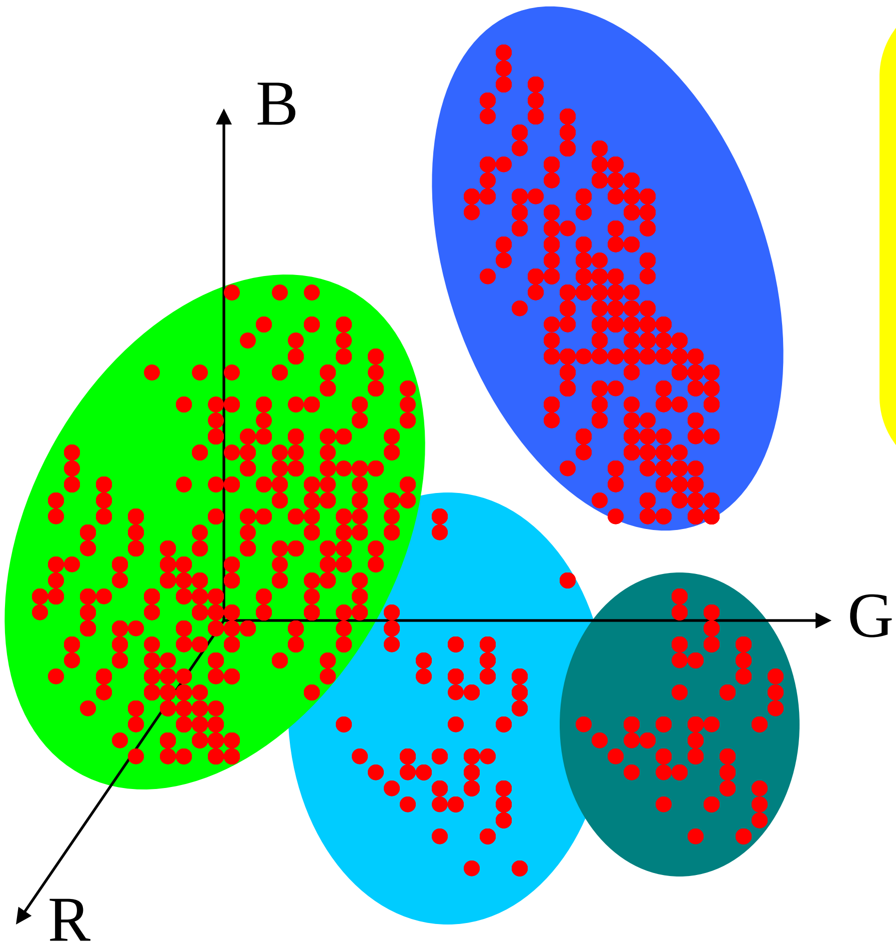
Motivating Problems

Map all pixels into the R, G, B space,
“clouds” of pixels are formed



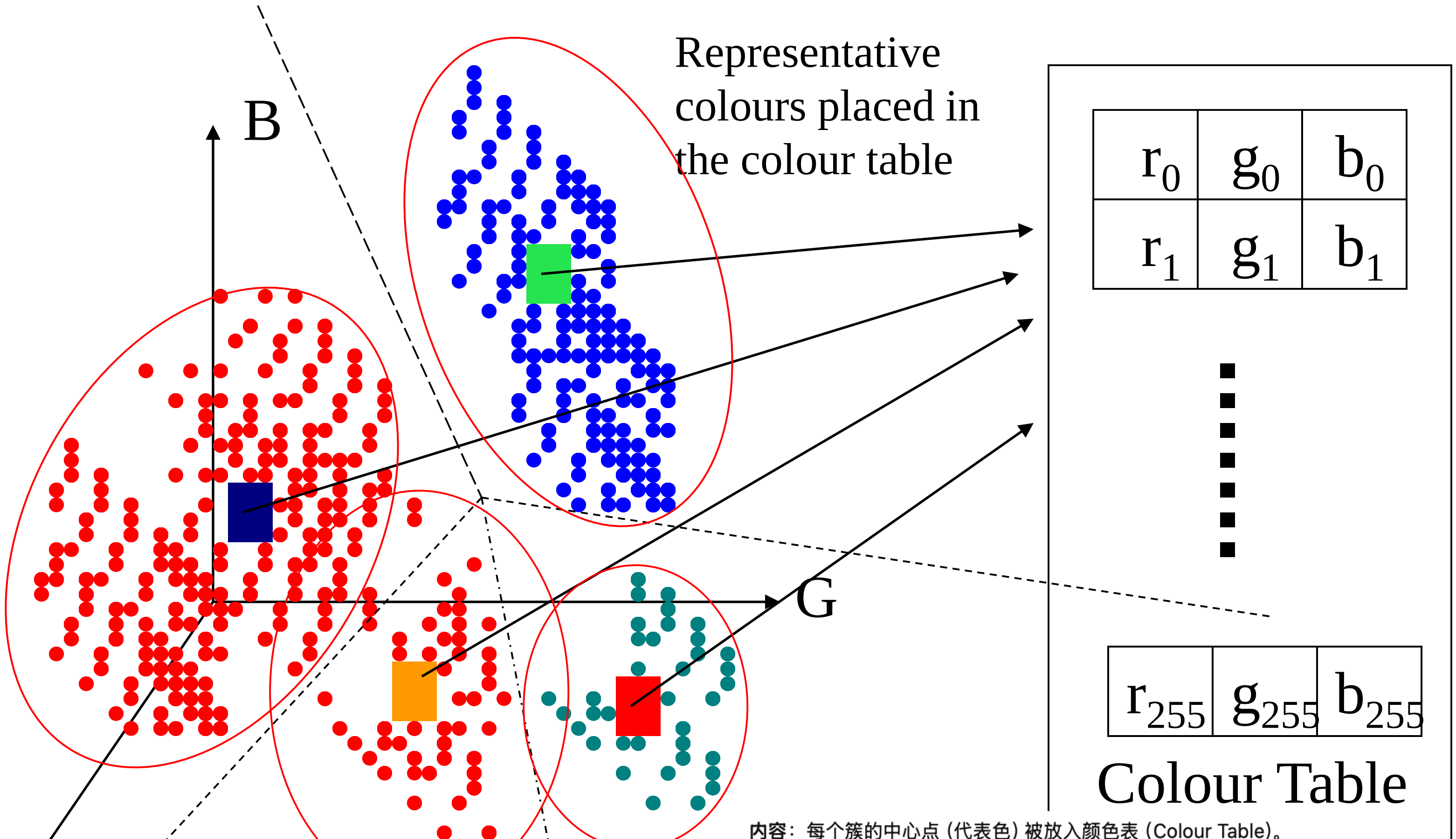
每个像素一个点 形成一个“点云”（clouds of pixels）。
不同颜色的像素会在空间中形成不同密集区域。
小结：图像颜色在 RGB 空间中呈现出若干自然分布的“颜色云”。

Motivating Problems



Group pixels that are close to each other, and replace them by one single colour

Motivating Problems



内容：每个簇的中心点（代表色）被放入颜色表（Colour Table）。

解释：

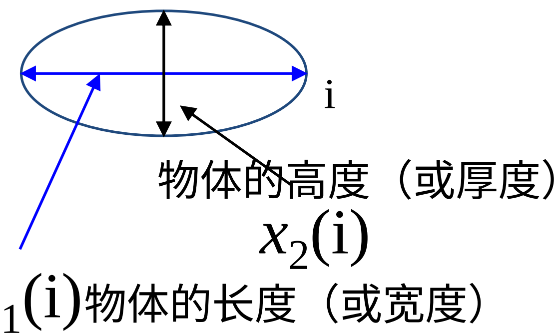
- 方块表示每个簇的“中心颜色”（簇均值）。
- 这些代表色就组成最终的 Colour Table。

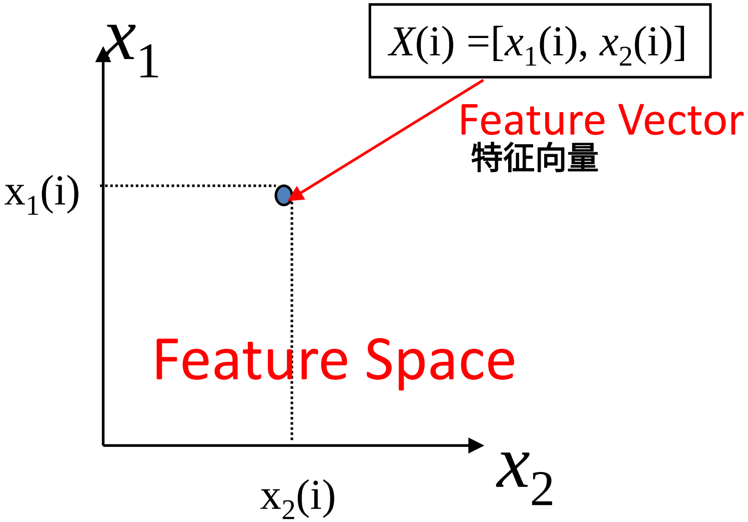
小结：

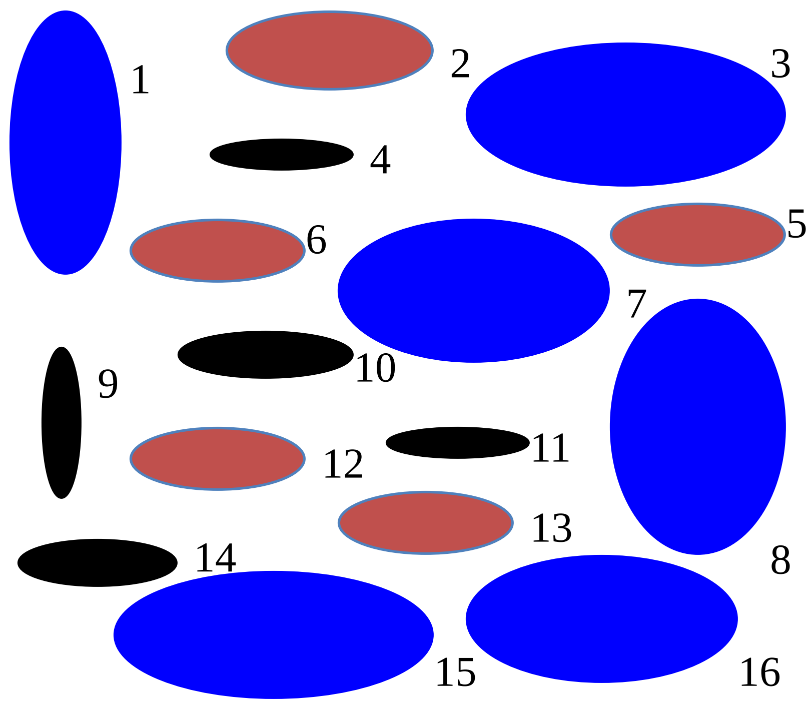
> 聚类得到的簇中心 = 颜色表中的代表色，用于图像压缩与重建。

Motivating Example

- Classify objects (Oranges, Potatoes) into large, middle, small sizes







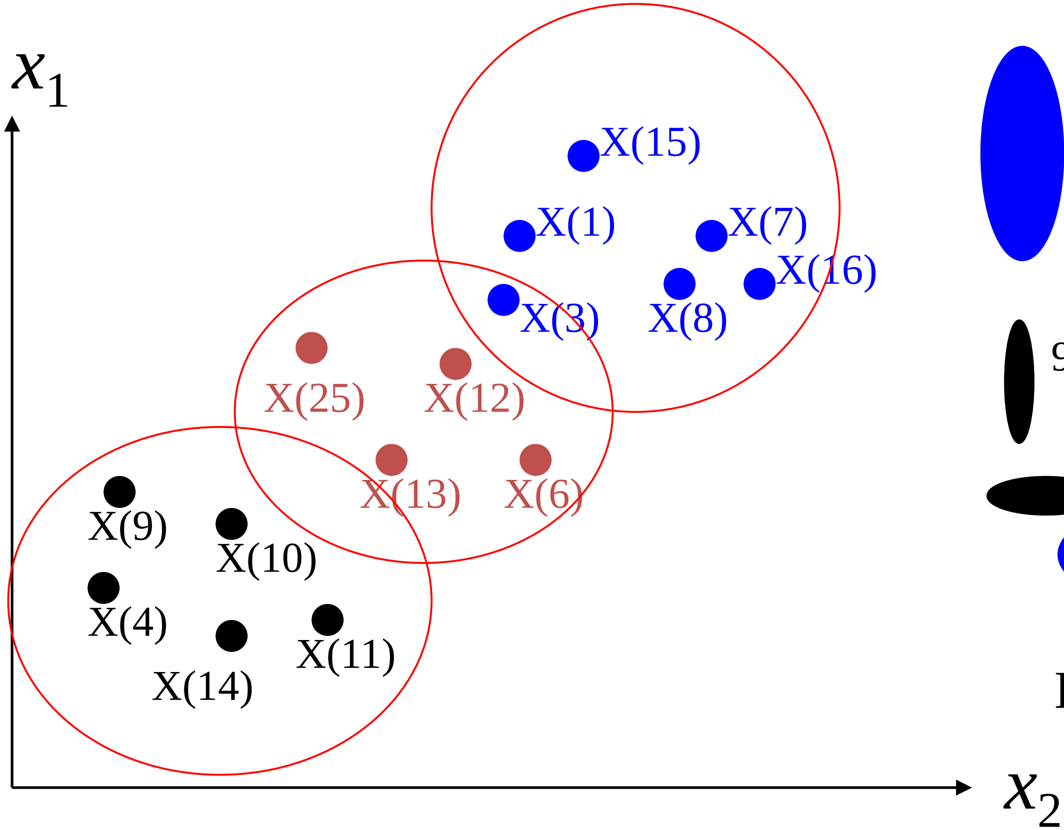
Elliptical blobs (objects)

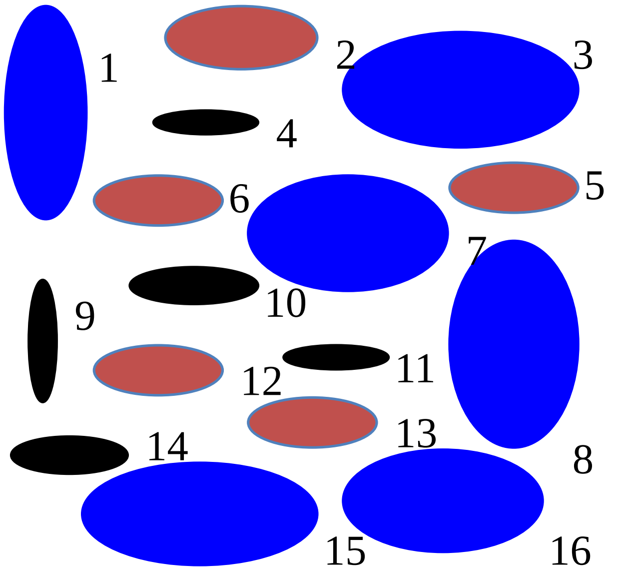
左下角坐标图展示的就是特征空间（Feature Space）：

- 横轴是 x_2，纵轴是 x_1；
- 每个点表示一个物体的大小特征。

Motivating Example

✪ From Objects to Feature Vectors to Points in the Feature Space





Elliptical blobs (objects)

Motivation of Clustering

聚类的目标是：同一簇内样本之间相似度高，不同簇之间相似度低。
关键思想：**簇内紧凑、簇间分离。**

- Patterns within a valid cluster are *more similar to each other* than they are to a pattern belonging to a different cluster.

聚类要处理的是**无标签数据**（unlabeled patterns），把它们分成有意义的组。
是一种 **data-driven method**，即：不依赖外部信息，只依据数据自身特征完成分组。

- In clustering, the problem is to group a given collection of *unlabeled patterns* into meaningful clusters. Clustering is data *driven method*, the clusters are obtained solely from the data.

- Clustering could be used in the field of pattern-analysis, grouping, decision-making, and machine-learning situations, including data mining, document retrieval, image segmentation

说明聚类的应用领域：
模式识别与分析（pattern analysis）数据挖掘（data mining）
文档检索（document retrieval）图像分割（image segmentation）
决策与群体划分（decision-making & grouping）

K-Means

- An algorithm for partitioning (or clustering) N data points into K disjoint subsets S_j containing N_j data points 把 N 个样本划分为 K 个不相交的簇（subset S, S, ··· S）。K由用户预设

  - Define, $X(i) = [x_1(i), x_2(i), \ldots, x_n(i)]$, i = 1, 2, ...N, as N data points 数据点

  - We want to cluster these N points into K subsets, or K clusters, where K is pre-set

  - For each cluster, we define $M(j) = [m_1(j), m_2(j), \ldots, m_n(j)]$, j=1, 2, ...K, as its prototype or cluster centroids 簇中心（质心）

  距离度量：即**欧氏距离**，衡量样本与簇中心的接近程度。

  - Define the distance between data point X(i) and cluster prototype M(j) as

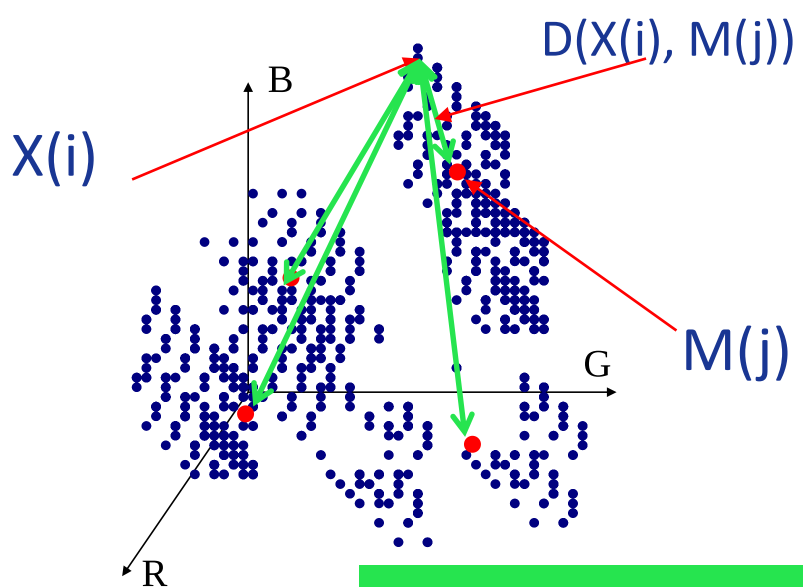
$$D\big(X(i), M(j)\big) = \left\| X(i) - M(j) \right\| = \sqrt{\sum_{l=1}^{n} \big(x_l(i) - m_l(j)\big)^2}$$

K-Means = “ 基于距离最小化的 K 类划分算法 ”；核心是**计算点与中心的欧氏距离**。

K-Means

- A data point X(i) is assigned to the jth cluster, C(j), $X(i) \in C(j)$, if following condition holds

$$D\big(X(i), M(j)\big) \leq D\big(X(i), M(l)\big) \quad \text{for all } l = 1, 2, \ldots, k$$



Minimum distance classifier

1. 点的归属规则

$X(i) \in C(j) \text{ if } D(X(i), M(j)) \leq D(X(i), M(l)), \ \forall l = 1K$

→ 意思是：把样本 $X(i)$ 分到与它最近的那个簇 $C(j)$。

→ 这就是“最小距离分类器 (Minimum distance classifier)”。

2. 图示说明

- 蓝色点：数据样本 $X(i)$
- 红色点：簇中心 $M(j)$
- 绿色箭头：样本到各中心的距离，选取最短的一条。
- 结果：不同点根据距离被划入不同簇。

🧠 小结

> 分配原则：每个样本归到最近的簇中心，这一规则是 K-Means 的核心步骤。

K-Means Algorithm

Step 1

从样本集中**任意选择 k 个初始簇中心**

✪ Arbitrarily choose from the given sample set k initial cluster centres,

$M^{(0)}(j) = [m^{(0)}_1(j), m^{(0)}_2(j), ,..., m^{(0)}_n(j)]\quad j = 1, 2, ..., K,$

e.g., the first K samples of the sample set
or can also be generated randomly

Set $t = 0$ (t is the iteration index)

- 通常做法：
  - 取前 K 个样本点；
  - **或随机选取** K 个点；
  - 或使用 k-means++ 更稳健。
- 设迭代计数 t = 0。

🧠 **小结**

> 第一步是**随机确定 K 个初始中心**。初始化方式不同可能导致最终结果差异。

K-Means Algorithm

Step 2

- ✪ Assign each of the samples $X(i) = [x_1(i), x_2(i), \ldots, x_n(i)]$, $i = 1, 2, \ldots N$, to one of the clusters according to the distance between the sample and the centre of the cluster:

$$X(i) \in C^{(t)}(j)$$
$$if \quad D\left(X(i), M^{(t)}(j)\right) \leq D\left(X(i), M^{(t)}(l)\right)$$
$$\text{for all } l = 1, 2, \ldots, k$$

💡 **内容说明**

- 对每个样本 X(i)，计算它与所有簇中心的距离：
$D(X(i), M^{(t)}(j))$

- **分配规则：**
$X(i) \in C^{(t)}(j) \quad D(X(i), M^{(t)}(j)) \leq D(X(i), M^{(t)}(l)), \ \forall\, l = 1, \ldots, K$

- 即把样本分给距离最近的中心。

🧠 **小结**

> 这一步把所有样本归入最近的簇 —— 叫 **“最小距离分类器”**。

K-Means Algorithm

Step 3

Update the cluster centres to get

$M^{(t+1)}(j) = [m^{(t+1)}_1(j), m^{(t+1)}_2(j), ,\dots, m^{(t+1)}_n(j)]$; $j = 1, 2, \dots, K$

according to

$$M^{(t+1)}(j) = \frac{1}{N_j^{(t)}} \sum_{X(i) \in C^{(t)}(j)} X(i)$$

$N^{(t)}_j$ is the number of samples in $C^{(t)}_j$

💡 **内容说明**

- 对每个簇 $C_{(j)}$，计算新的质心（均值）：

$$M^{(t+1)}(j) = \frac{1}{N_j^{(t)}} \sum_{X(i) \in C^{(t)}(j)} X(i)$$

- 其中 $N_j^{(t)}$ 是当前簇 j 中样本的数量。
- 新的 M 替换旧的 M 用于下一次迭代。

🧠 **小结**

> 每一轮更新中心点为簇内样本的平均值，使中心向“密集区”移动。

K-Means Algorithm

Step 4

– Calculate the error of approximation

$$E(t)=\sum_{j=1}^{K}\sum_{X(i)\in C^{(t)}(j)}\left\|X(i)-M^{(t)}(j)\right\|$$

💡 **内容说明**

- 计算当前迭代的总误差 E(t)：

$$E(t)=\sum_{j=1}^{K}\sum_{X(i)\in C^{(t)}(j)}\|X(i)-M^{(t)}(j)\|$$

- 该值衡量簇内紧凑程度（簇内平方误差 SSE）。
- 随迭代次数 t 增加， E(t) 应逐步减小。

🧠 **小结**

> 误差 E(t) 表示聚类效果；目标是**最小化簇内距离和。**

K-Means Algorithm

Step 5

- If the terminating criterion is met, then stop, otherwise

  Set t = t+1

  Go to Step 2.

💡 **内容说明**

- 若满足停止条件，则终止；
- 否则令 t = t + 1，返回 Step 2 继续迭代。

🧠 **小结**

> 不断循环“分配 → 更新 → 计算误差”，直到满足终止标准。

K-Means Algorithm

Stopping criterions

– The K-means algorithm can be stopped based on following criterions

1. The errors do not change significantly in two consecutive epochs **误差收敛**

   $|E(t)-E(t-1)|<\varepsilon$, where ε is some preset small value 连续两轮误差变化很小

2. No further change in the assignment of the data points to clusters in two consecutive epochs. **簇分配稳定**：两轮迭代中样本归属不再改变。

3. It can also stop after a fixed number of epochs regardless of the error
   最大迭代数限制：达到预设迭代次数后强制停止。

小结
当误差几乎不变或簇稳定时算法收敛；否则继续迭代。

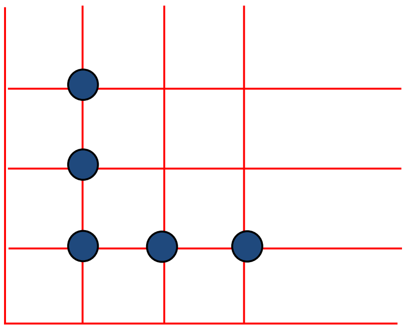
K-Means Algorithm

A worked example to see how it works exactly

Five 2-dimensional data points:

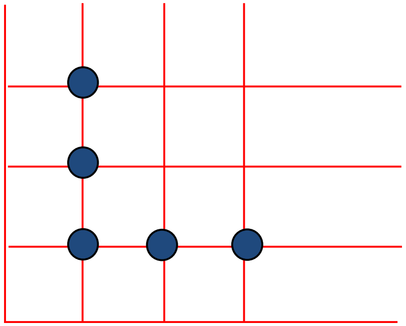
(1, 1), (2, 1), (3, 1), (1, 2), (1, 3)

Cluster them into two clusters and find the cluster centres



K-Means Algorithm

A worked example to see how it works exactly



(1) Euclidean distance to $m_1^0(1,2)$	Euclidean distance to $m_2^0(3,1)$	
$\sqrt{(1-1)^2+(1-2)^2}=1$	$\sqrt{(1-3)^2+(1-1)^2}=2$	$\therefore C_1$
$\sqrt{(2-1)^2+(1-2)^2}=\sqrt{2}=1.41$	$\sqrt{(2-3)^2+(1-1)^2}=1$	$\therefore C_2$
$\sqrt{(3-1)^2+(1-2)^2}=\sqrt{5}=2.24$	$\sqrt{(3-3)^2+(1-1)^2}=0$	$\therefore C_2$
$\sqrt{(1-1)^2+(2-2)^2}=0$	$\sqrt{(1-3)^2+(2-1)^2}=\sqrt{5}=2.24$	$\therefore C_1$
$\sqrt{(1-1)^2+(3-2)^2}=1$	$\sqrt{(1-3)^2+(3-1)^2}=\sqrt{8}=2.83$	$\therefore C_1$

C_1 class: $(1,1), (1,2), (1,3) \Longrightarrow m_1^{(0+1)}$:

$$\frac{1}{3}\sum_{i=1}^{3} x_i = \frac{1+1+1}{3} = 1$$

$$\frac{1}{3}\sum_{i=1}^{3} y_i = \frac{1+2+3}{3} = 2$$

C_2 class: $(2,1), (3,1) \Longrightarrow m_2^{(0+1)}$:

$$\frac{1}{2}\sum_{i=1}^{2} x_i = \frac{2+3}{2} = 2.5$$

$$\frac{1}{2}\sum_{i=1}^{2} y_i = \frac{1+1}{2} = 1$$

(2) Euclidean distance to $m_1^1(1,2)$	Euclidean distance to $m_2^1(2.5,1)$	
1	$\sqrt{(1-2.5)^2+(1-1)^2}=1.5$	$\therefore C_1$
$\sqrt{2}=1.41$	$\sqrt{(2-2.5)^2+(1-1)^2}=0.5$	$\therefore C_2$
$\sqrt{5}=2.24$	$\sqrt{(3-2.5)^2+(1-1)^2}=0.5$	$\therefore C_2$
0	$\sqrt{(1-2.5)^2+(2-1)^2}=\sqrt{3.25}=1.80$	$\therefore C_1$
1	$\sqrt{(1-2.5)^2+(3-1)^2}=\sqrt{6.25}=2.5$	$\therefore C_1$

$\therefore C_1$ class: $(1,1), (1,2), (1,3)$

C_2 class: $(2,1), (3,1)$

K-Means Algorithm

- What is the algorithm doing exactly?
  - It tries to find the centre vectors M(j)'s that optimize the following cost function

$$E = \sum_{j=1}^{K} \sum_{X(i) \in C(j)} \left\| X(i) - M(j) \right\|$$

K-Means Algorithm

- Some remarks
  - Is a gradient descent algorithm, trying to minimize a cost function *E*
  - In general, the algorithm does not achieve a global minimum of E over the assignments
  - Sensitive to initial choice of cluster centers. Different starting cluster centroids may lead to different solution
  - Is a popular method, many more advanced methods derived from this simple algorithm

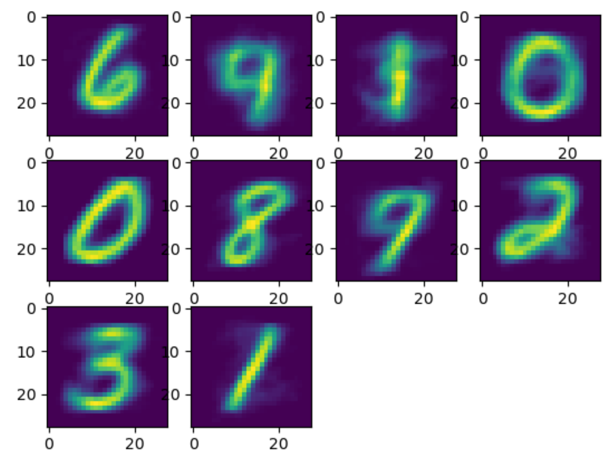
- K-Means ≈ **坐标下降**最小化簇内平方误差；
- **非凸** → 只保局部最优，**对初始化敏感**；
- 用 **k-means++ + 多次重启 + 标准化 + 去离群** 提升稳定性；
- 作为基线很强，但遇到**非球形/密度不均/离群多**的数据要考虑替代算法。

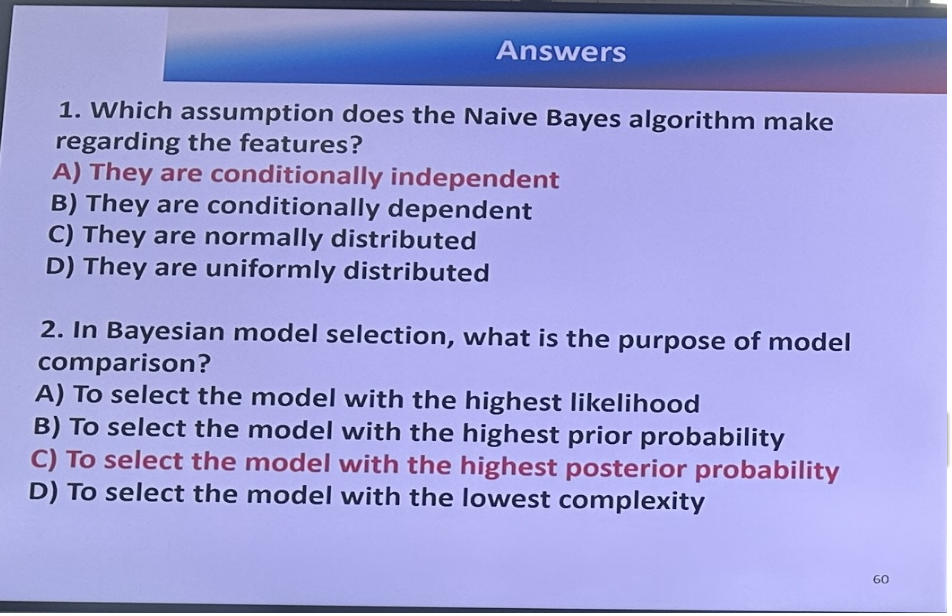
K-Means Clustering Coding

- Use KMeans from sklearn.cluster to do K-Means clustering

  See the following as an example:

```
from sklearn.cluster import KMeans
kmeans = KMeans(n_clusters=10).fit(X_small)
kmeans.cluster_centers_
kmeans.labels_ = y_small
kmeans.predit(X_test)
```





Answers

1. Which assumption does the Naive Bayes algorithm make regarding the features?

A) They are conditionally independent
B) They are conditionally dependent
C) They are normally distributed
D) They are uniformly distributed

2. In Bayesian model selection, what is the purpose of model comparison?

A) To select the model with the highest likelihood
B) To select the model with the highest prior probability
C) To select the model with the highest posterior probability
D) To select the model with the lowest complexity

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estions?